

Databricks Introduction

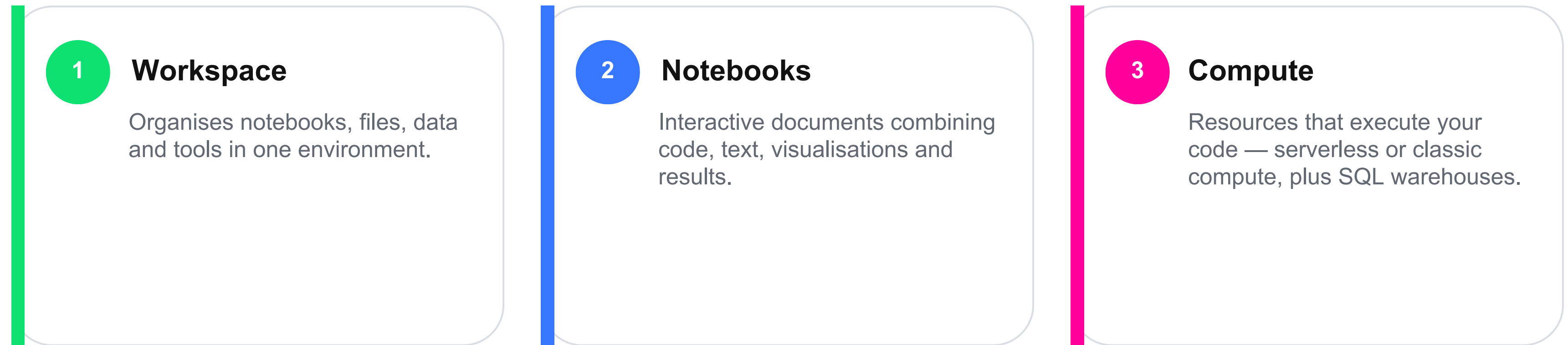


databricks

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What is Databricks?

Databricks is a cloud-based data platform that unifies data engineering, analytics, and artificial intelligence into a single workspace. Think of it as a workspace where we develop, run, and collaborate on data workloads.



You focus on the data work → Databricks manages the environment around it.

Databricks notebooks

Cells make the workflow interactive: write → run → inspect → change.

my_first_notebook

MARKDOWN

Explore customer transactions

PYTHON

df = spark.read.csv("transactions.csv")

PYTHON

display(df.groupBy("category").count())

output

category	count
A	12,481
B	9,205

Interactive

Run one cell or many.

Multi-language

Python, SQL, Scala and R.

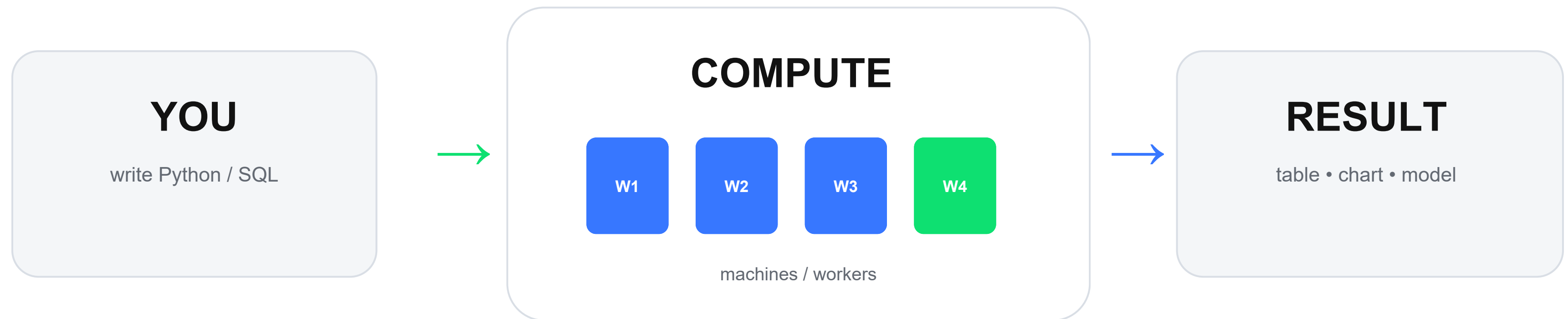
Visual

Inspect tables and charts immediately.

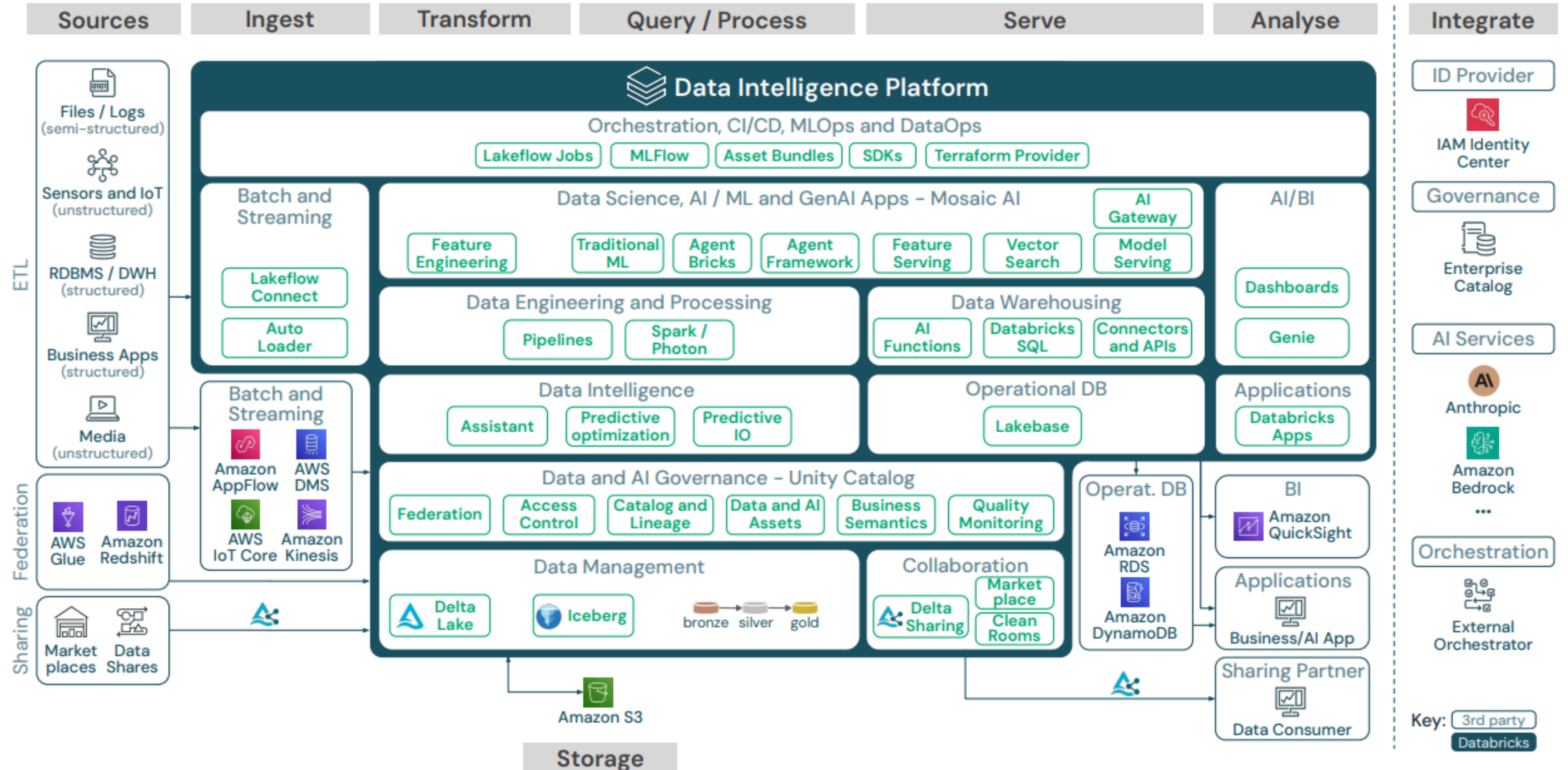
Compute: where does the code run?

When you run a Python or Spark command in Databricks, the code is not executed directly by your browser. The notebook is connected to a compute resource (cluster) that runs the code for you.

The compute resource provides the CPU, memory, and other resources needed to execute your code. For Spark applications, the work can be distributed across multiple machines, allowing Databricks to process large datasets efficiently.



Databricks Data Intelligence Platform



Why Apache Spark?

When the data or computation becomes too large for a single machine, Spark can distribute the work across multiple machines in the Databricks compute cluster, allowing large-scale data processing to be performed efficiently.

ONE MACHINE



all data + all work

Can be perfectly fine for many datasets.

DISTRIBUTED WITH SPARK



data / tasks split across workers

The goal is not “use Spark for everything”; it is to scale processing when needed.

Pandas vs Spark: when do we scale?

The important difference is where the computation happens.

pandas

PySpark / Spark

Data

Usually fits in one machine's memory

Designed for distributed datasets

Execution

Local Python process

Spark engine across compute resources

Scale

Simple and fast for many small/medium tasks

Useful when data/work exceeds local capacity

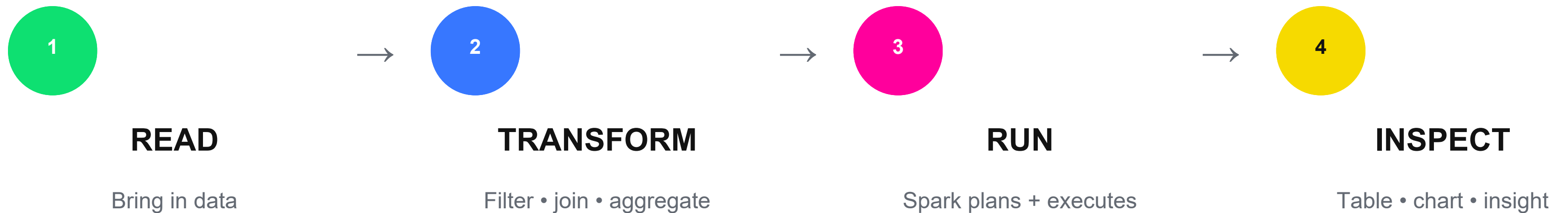
Mindset

Work with local objects

Think DataFrames + distributed execution

Rule of thumb → start simple; use distributed processing when scale demands it.

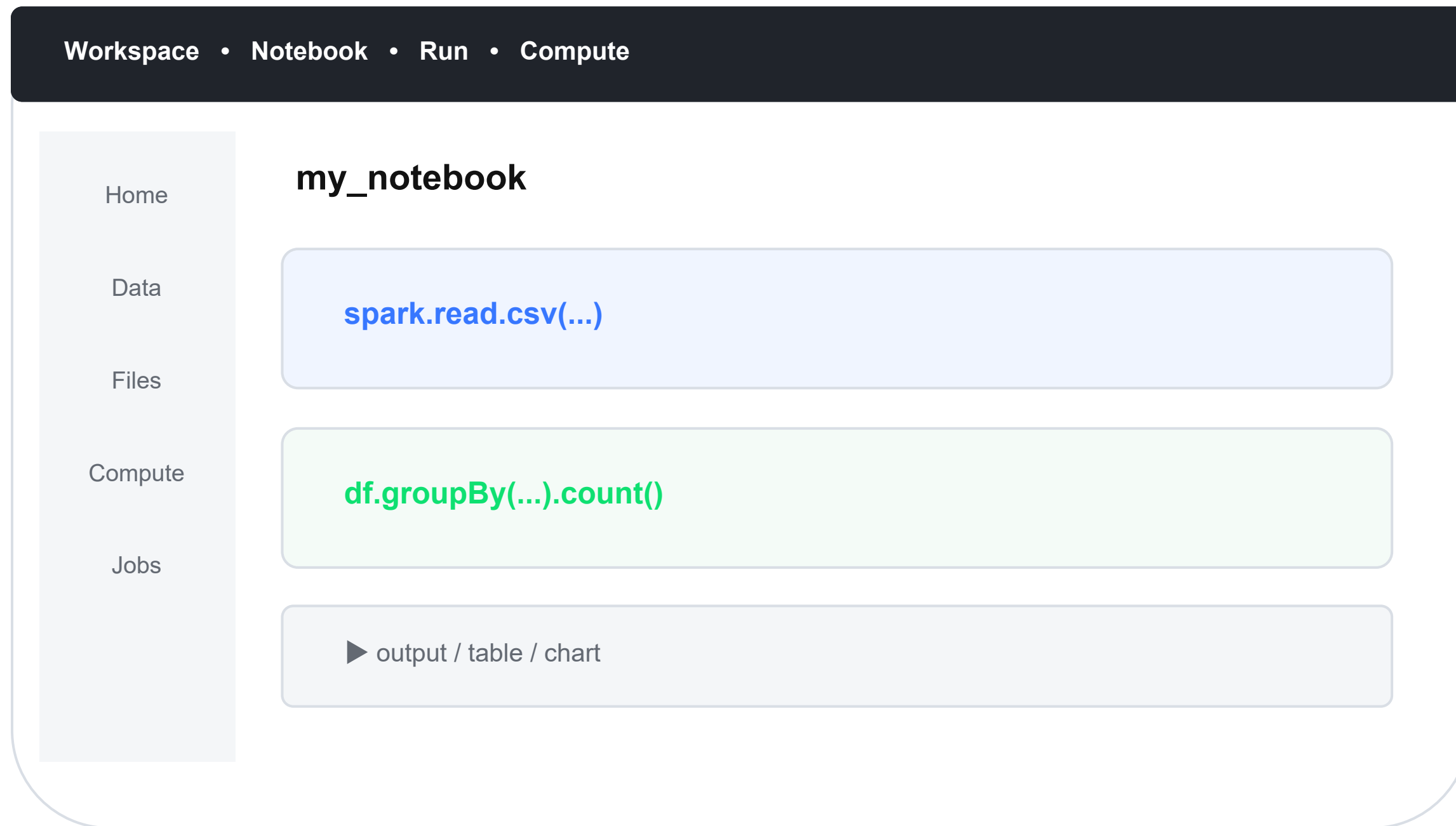
The mental model for today



Key idea

You describe the transformation; Spark handles the distributed execution plan.

Databricks notebook: what should you recognise?



Notebook

Where you write and run cells.

Compute selector

Where the notebook runs.

Output

Where you inspect results.

Three things to remember

- 1 Databricks** The environment where we build and run data workloads.
- 2 Notebook** The interactive document where code, text and results live.
- 3 Spark** The engine that enables distributed data processing.

Next: open a notebook → connect compute → load data → run your first Spark DataFrame.

Thank

you!

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